

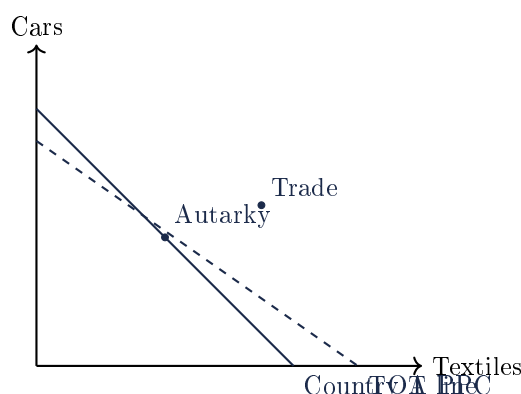
Document 21. International Trade & Comparative Advantage

Topic overview. Comparative advantage explains why nations gain from trade even when one country is absolutely more productive. Singapore's open-economy strategy hinges on this principle.

Section A. Definitions

Absolute advantage	Country produces more of a good per unit of input.
Comparative advantage	Country produces a good at a lower opportunity cost.
Terms of trade (TOT)	$\frac{\text{Index of export prices}}{\text{Index of import prices}} \times 100$.
FTAs	Free Trade Agreements; bilateral or multilateral preferential tariff arrangements.
Trade creation	FTA shifts imports from high-cost domestic to low-cost FTA partner → welfare gain.
Trade diversion	FTA shifts imports from lower-cost non-member to higher-cost member → welfare loss.

Section A1. Diagram — Gains from Trade (PPC)



Caption. Specialising according to comparative advantage and trading along the TOT line allows Country A to consume *outside* its PPC.

Section B. Question Type 1: Law of comparative advantage

- Step 1.** Set up a two-good, two-country table with output per worker hour.
- Step 2.** Compute opportunity cost of each good in each country.
- Step 3.** Identify the lower-opportunity-cost producer for each good.
- Step 4.** Show that trade at a TOT between the two opportunity-cost ratios benefits both.
- Step 5.** Conclude with the gains-from-trade triangle (output gains + consumption gains).

Section C. Question Type 2: Gains from trade

- **Consumer gain.** Wider variety, lower prices, higher quality.
- **Producer gain.** Larger market → scale economies.
- **Dynamic gain.** Technology transfer, innovation, competitive discipline.

- **Macro gain.** Higher output, employment, tax revenue.

Section D. Question Type 3: Costs of free trade

- Structural unemployment in import-competing sectors.
- Over-reliance on foreign demand; vulnerability to global cycles.
- Externalities: emissions from shipping, environmental damage from extraction.
- Income inequality between trade winners and losers.
- National security risks (concentration in strategic imports).

Section E. Question Type 4: Trade creation vs diversion (FTAs)

Step 1. Define trade creation and trade diversion.

Step 2. For trade creation, draw the SS-DD diagram with world price and tariff; show gain when tariff eliminated for FTA partner.

Step 3. For diversion, show how a preferential tariff diverts trade from a lower-cost non-member to a higher-cost member, raising prices for consumers.

Step 4. Conclude that FTAs are welfare-enhancing only if creation > diversion.

Section F. Singapore Context

- Trade-to-GDP ratio >300%; export-oriented.
- Singapore's comparative advantage: high value-added manufacturing (electronics, pharmaceuticals), financial services, transshipment / logistics.
- 27+ FTAs including **CPTPP**, **RCEP**, **USSFTA**, **EUSFTA**, **CSFTA** (China).
- Singapore's FTA strategy emphasises trade creation while minimising diversion (covers diverse partners).

Section G. Worked Example

Worked Example 1: A vs B with absolute and comparative advantage

Country A: 1 worker-hour makes 4 cars or 8 textiles. Country B: 1 worker-hour makes 1 car or 4 textiles. Opportunity cost of 1 car: A = 2 textiles, B = 4 textiles. A has CA in cars. Opportunity cost of 1 textile: A = 0.5 cars, B = 0.25 cars. B has CA in textiles. At any TOT between 2 and 4 textiles per car (e.g. 3 textiles per car), both gain. A trades cars for textiles, getting more textiles than at home. B trades textiles for cars, getting more cars than at home. Both consume outside their respective PPCs. *Conclusion:* comparative advantage governs trade even when A has absolute advantage in both.